

# ARITRO DUTTA

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**Work Rights: Full Working Rights in Australia (Subclass 485 Graduate Visa - Valid to Jan 2028, No Work Restrictions)**

## PROFESSIONAL SUMMARY

**Software Engineering (Honours)** graduate from **Monash University** with production experience in AI development, computer vision, full-stack web engineering, and data pipelines. Proven track record of designing and deploying production RAG architectures, custom PyTorch vision models, and full-stack React/FastAPI applications. Proficient in Python, TypeScript, React, SQL, and Microsoft Azure, leveraging modern AI development tools (Claude, Cursor) to deliver reliable, high-impact software solutions. Available immediately.

## EDUCATION

### Bachelor of Software Engineering (Honours) | Monash University

Dec 2025

Melbourne, VIC

- **Grade:** Second Class Honours Division A (H2A)
- **Scholarship:** Faculty of Engineering International Undergraduate Excellence Scholarship
- **High Distinction Coursework:** Software Security (90%), Final Year Project A (85%), Software Architecture & Design (82%), Software Engineering Practice (80%)

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, JavaScript, TypeScript, React, FastAPI, Flask, SQL, PostgreSQL, MongoDB, PHP, HTML/CSS

**AI & Machine Learning:** PyTorch, RAG Pipelines, LangChain, FAISS, YOLOv9, OpenCV, LLM Integration, Prompt Engineering, Generative AI

**Cloud & Infrastructure:** Microsoft Azure (Azure AI, Blob Storage), REST APIs, OAuth 2.0, Supabase, Git, GitHub, CI/CD

**Data Engineering & Automation:** Pandas, NumPy, Data Pipelines, Playwright, Tableau, Automated Workflows

**Development Tools:** Claude, Cursor, GitHub Copilot, Gemini API, OpenAI API, Linux

## WORK EXPERIENCE

### AI Developer | QEERI.AI

Mar 2024 - Oct 2024

Melbourne, VIC (Industry Placement via Monash University)

- Engineered and deployed a production Generative AI & RAG platform using **Python, LangChain, and FAISS**; conducted systematic A/B prompt testing that **improved retrieval accuracy by 40%** over baseline models.
- Architected a full-stack Web application using **FastAPI, React, and Azure AI Services**, maintaining **98% uptime** for 100+ active users and managing cloud storage via **Azure Blob Storage**.
- Built automated ETL and document processing pipelines for 500+ documents, **slashing manual processing effort by 95%** and increasing data ingestion efficiency.
- Collaborated closely with cross-functional stakeholders to translate commercial requirements into technical features, conducting code reviews and iterative product updates.

### Computer Vision Engineer | CSIRO

Mar 2023 - Oct 2023

Melbourne, VIC (Industry Placement via Monash University)

- Trained and evaluated custom **YOLOv9 object detection models in PyTorch**, achieving 92% mean average precision (mAP) across 50+ target categories for a **government research deployment**.
- Profiled and optimized the execution pipeline in Python, eliminating CPU/GPU bottlenecks to cut system latency by 15ms for real-time edge processing.
- **Diagnosed and resolved hardware/software integration issues through systematic root-cause analysis**, reducing hardware sync error rates below 1%.
- Produced technical documentation and presented findings across interdisciplinary research and engineering teams.

### Full Stack Web Developer | Creation Export

Dec 2021 - Feb 2022 | Ongoing 2024 - 2025

- **Designed, built, and launched a full-stack e-commerce platform** using JavaScript, PHP, REST APIs, and OAuth 2.0, providing ongoing feature development and platform maintenance.
- Modeled and implemented the relational **SQL database schema** to handle concurrent product listings, and user profiles securely.

### Avionics Software Developer | Monash Uncrewed Aerial Systems

Apr 2022 - Feb 2024

Melbourne, VIC

- Developed autonomous flight control software within a 50-member engineering team, contributing to a **15% increase in flight mission success rate**.
- Resolved communication and telemetry protocol errors across 10+ interconnected electronic systems, **reducing overall error rates by 25%**.

Platform achieved Top 10% global ranking at The New Flying Competition in Hamburg, Germany, after I had moved on from the team in February 2024.

PROJECTS

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ApplyBot | Python, Node.js, Google Gemini API, Playwright, React, Claude2025 - 2026

- Built an automated AI job application agent integrating **Google Gemini** for tailorable cover letter generation, **Playwright** for browser automation, and **Adzuna API** for job discovery.
- Developed a **React analytics dashboard** providing real-time metrics on application statuses and success rates.

Universal Book | React, TypeScript, Supabase, PostgreSQL, REST APIs2025

*Monash Honours Capstone Project*

- Architected a real-time collaborative web application end-to-end, designing the relational **PostgreSQL** schema and custom **RESTful API endpoints**.

Logistics & Resource Optimisation Engine | Python, Graph Theory, Network Flow2025

- Formulated a **constrained multi-source transit allocation system** by mapping spatial locations and transportation corridors into a directed weighted graph, optimizing reachability metrics under bounded distance thresholds.
- Solved a non-linear capacity allocation problem by transforming resource constraints into a multi-sink flow network, **achieving optimal agent-to-capacity matching within hard lower and upper bounds in Big O time complexity and auxiliary space**.

Music Pattern Analytics Engine | Python, Advanced Data Structures, String Algorithms

- Engineered a specialized multi-string prefix/suffix index tree to analyse contiguous sequential motifs across multi-song dynamic datasets, enabling pattern retrieval for targeted sequence lengths.
- Integrated dynamic pitch-transposition invariance logic into the indexing pipeline, enabling real-time detection of frequency-shifted structural equivalences and recurring melodic motifs across distinct sequence sets.

CERTIFICATIONS

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- **HackerRank Software Engineer Certificate**
  - **100 Days of Code: Complete Python Pro Bootcamp - Udemy**
  - **ANZ Cyber Security Management Virtual Experience**
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